ENTERPRISE MANAGER DATABASE REPLAY EXECUTIVE SUMMARY

sh_upgradetest_01

sh_upgradetest_01 finished with status COMPLETED. Replay DB time improved by 57.6% versus capture. Divergence is LOW at 2.79% of calls. Overall verdict: good.

PASS Bottom line: good

RISK RATING

Low

REPLAY DATABASE

DB26(23.26.1.0.0)

CAPTURE DATABASE

SDB2(19.3.0.0.0)

Executive Summary

The replay completed successfully. This was an upgrade test from **19.3.0.0.0** to **23.26.1.0.0**. Overall DB Time is lower in replay, which is the primary positive performance signal (-**57.6%**). Replay has fewer CPU cores than capture (2 vs 1), which weakens comparability and can explain slower replay. Replay also has lower physical memory than capture, which can further distort the comparison. Replay divergence is **2.79%**, which supports a broadly reliable replay.

Test Outcome

- **Status:** COMPLETED

- **Was the test valid?** Yes - Replay completed and the core comparability checks do not show a major fidelity failure.

Test Objective

- **Change being tested:** Version upgrade
- **Reason:** Capture ran on **19.3.0.0.0** and replay ran on **23.26.1.0.0**.

Performance Assessment

- **Overall:** Good
- **DB Time:** Improved
- **User Calls:** Similar
- **Divergence:** Good
- **Session completion:** Stable

Key Findings

- **Replay outcome:** Replay completed successfully.
- **DB Time:** DB Time is improved in replay (7572.39 seconds -> 3211.44 seconds, -**57.6%**).
- **User Calls:** User Calls are similar between capture and replay (110589 -> 110589, **0.0%** difference).
- **Finished Replay Sessions:** Finished Replay Sessions stayed broadly consistent (79 -> 79).
- **Replay divergence:** Replay divergence is **LOW** at **2.79%**, which is within the **good** range for comparability.

- **SELECT result differences:** 3080 SELECT calls fetched a different number of rows during replay.
- **Platform:** Platform appears unchanged (Linux x86 64-bit).
- **CPU configuration: CPU cores decreased from 2 to 1**, which can degrade performance.
- **Memory configuration:** Physical memory decreased from 15.33 G to 15.17 G.
- **Parameter change:** Important parameter compatible changed from 19.0.0 to 23.6.0.
- **Parameter change:** Optimizer parameter pga_aggregate_target changed from 1.53 G to 1.52 G.
- **Parameter change:** Optimizer parameter optimizer_features_enable changed from 19.1.0 to 23.1.0.
- **Parameter change:** Memory parameter sga_max_size changed from 4.61 G to 4.56 G.
- **CPU throttling:** `resmgr:cpu quantum` indicates replay is seeing CPU throttling or scheduling pressure.
- **Parsing overhead:** Parse time share rose from **19.33%** to **30.53%**. Total parses increased from 54,663 to 96,786.

Likely Causes

- Replay environment is CPU-downgraded versus capture.
- Replay environment has less physical memory than capture.
- Configuration differences between environments may be contributing to the performance change.
- Resource Manager or CPU pressure is constraining replay throughput.
- Higher parse pressure or literal SQL usage is adding overhead during replay.

Final Verdict

✅ **Successful test result**

Replay status and fidelity checks are sufficient for a directional conclusion.

Why this result was assigned:

- Replay completed and the core comparability checks do not show a major fidelity failure.
- Capture ran on **19.3.0.0.0** and replay ran on **23.26.1.0.0**.
- Performance assessment for DB Time: **Improved**.
- Divergence assessment: **Good**.

Diagnostic Notes

Moderate

Replay divergence is present

Issue: LOW divergence with 2.79% of calls diverging.

Likely cause: The main divergence signal comes from SELECT row-count differences, which often indicates data drift, optimizer plan changes, or environmental differences between capture and replay.

Recommendation: Validate schema statistics, application data parity, initialization parameters, and replay setup before drawing strong performance conclusions.

Low

Replay completed faster overall

Issue: Database Time decreased by 57.59% in replay.

Likely cause: The replay workload consumed less total DB time than capture, suggesting faster completion or lower effective concurrency.

Recommendation: Treat this as a positive signal, but confirm the lower DB time is not being masked by divergence or reduced workload fidelity.

Low

Replay shifted toward CPU-heavy execution

Issue: CPU time increased by 3.8% and now represents 26.12% of replay DB time.

Likely cause: The target system is spending a larger share of time on CPU, which can happen after I/O bottlenecks are reduced, when SQL plans change, or when parsing and metadata work increase.

Recommendation: Review top SQL, execution plans, and CPU headroom to ensure the improvement does not create a new CPU ceiling.

Low

Concurrency waits remain visible in replay

Issue: latch: shared pool accounts for 0.68% of replay DB time with 5076 waits.

Likely cause: Library cache and row cache contention usually point to parse pressure, shared pool churn, metadata contention, or repeated object invalidation.

Recommendation: Reduce hard parsing, check shared pool sizing, stabilize SQL reuse, and review dictionary-heavy operations during replay.

Evidence Snapshot

Metric	Value
Replay status	COMPLETED
Replay divergence	LOW (2.79%)
Database Time change	-57.59%
CPU Time change	3.8%
Replay average active sessions	5.53
AWR top replay wait	resmgr:cpu quantum
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Prototype Verdict Criteria

This section explains the prototype-only scoring logic used to derive the Bottom Line and Risk Rating. It is intended as reviewer guidance and methodology, not as part of the replay performance findings themselves.

How Bottom Line Is Defined

This prototype uses a rule-based scoring approach, not an LLM, to assign the Bottom Line. Reviewers can use the table below to see the explicit threshold logic.

Outcome	Prototype Rule
PASS / good	Assigned when Database Time improves materially, using DB Time change <= -**20%**, divergence stays below **3%**, Resource Manager CPU throttling stays below **10%**, and replay CPU capacity is not lower than capture.
WARN / mixed	Assigned when the result is inconclusive or partially positive: for example DB Time improves by less than **20%**, DB Time regression is **10%** or less, or comparability is weakened by throttling, CPU mismatch, or non-zero divergence.
FAIL / bad	Assigned when Database Time regresses materially, using DB Time change > +**10%**, or when the replay shows clear evidence that performance is worse in the test environment.

How Risk Rating Is Defined

Risk Rating is a separate roll-up of diagnostic findings. Each finding is assigned a severity and converted into a score:

- **High finding** = 35 points
- **Moderate finding** = 20 points
- **Low finding** = 8 points

The total score is then mapped to the displayed Risk Rating:

Risk Rating	Prototype Rule
Low	Total finding severity score below 60. Usually means only a few low or moderate issues were detected.
Moderate	Total finding severity score from 60 to 84. Usually means multiple moderate issues or one high issue plus other concerns were detected.
High	Total finding severity score 85 or above. Usually means several high-impact issues were detected in the replay analysis.